

Modern Quantum Mechanics J.J. Sakurai
Solution Manual

Vyas Mokal

January 20, 2026

Contents

1	Quantum Dynamics	5
1.1	Problem 2.1	5
1.2	Problem 2.18	6

Chapter 1

Quantum Dynamics

1.1 Problem 2.1

$$H = -\frac{eB}{mc}S_z$$

$$\begin{aligned}\frac{dA}{dt} &= \frac{1}{i\hbar}[A, H] + \frac{\partial A}{\partial t} \\ \frac{dS_x}{dt} &= \frac{1}{i\hbar}[S_x, \omega S_z] \\ &= \frac{\omega}{i\hbar}[S_x, S_z] \\ &= \frac{\omega}{i\hbar}i\hbar(-S_y)\end{aligned}$$

$$\dot{S}_x = \omega S_y \quad (1.1)$$

Similarly,

$$\frac{dS_y}{dt} = \frac{1}{i\hbar}\omega[S_y, S_z]$$

$$\dot{S}_y = \omega S_x \quad (1.2)$$

Now, $\ddot{S}_x = -\omega\dot{S}_y = -\omega(\omega S_x)$

$$\ddot{S}_x + \omega^2 S_x = 0 \quad (1.3)$$

$$S_x(t) = A \cos \omega t + B \sin \omega t \quad (1.4)$$

$\ddot{S}_y = \omega\dot{S}_x = \omega(-\omega S_y)$

$$\ddot{S}_y + \omega^2 S_y = 0 \quad (1.5)$$

$$S_y(t) = C \cos \omega t + D \sin \omega t \quad (1.6)$$

using initial conditions at $t = 0$

$$S_x(0) = S_x \quad S_y(0) = S_y$$

we can write,

$$S_x(t) = S_x \cos \omega t + B \sin \omega t \quad (1.7)$$

$$S_y(t) = S_y \cos \omega t + D \sin \omega t \quad (1.8)$$

But from (1.1)

$$-\omega S_x \sin \omega t + \omega B \cos \omega t = -\omega S_y \cos \omega t - \omega D \sin \omega t$$

$$B = -S_y \quad D = S_x$$

$$\boxed{S_x(t) = S_x \cos \omega t - S_y \sin \omega t} \quad (1.9)$$

$$\boxed{S_y(t) = S_y \cos \omega t + S_x \sin \omega t} \quad (1.10)$$

$$\frac{dS_z}{dt} = \frac{\omega}{i\hbar} [S_z, S_z] = 0$$

$$S_z(t) = C$$

at $t=0$ $S_z(0) = S_z$

$$\boxed{S_z(t) = S_z} \quad (1.11)$$

1.2 Problem 2.18

Ground state of 1-D Simple Harmonic Oscillator is given by

$$\psi_0(x, t) = \left(\frac{m\omega}{\pi\hbar}\right)^{\frac{1}{4}} \exp\left(-\frac{m\omega x^2}{2\hbar} - i\frac{E_0 t}{\hbar}\right) \quad (1.12)$$

with $E_0 = \frac{\hbar\omega}{2}$. Now the correlation function

$$\begin{aligned} C(t) &= \langle x(t)x(0) \rangle \\ &= \int_{-\infty}^{\infty} \left(\frac{m\omega}{\pi\hbar}\right)^{\frac{1}{2}} \exp\left(\frac{m\omega x^2}{\hbar} + i\frac{\omega t}{2}\right) dx \\ &= e^{\frac{i\omega t}{2}} \end{aligned}$$

$$\boxed{C(t) = e^{\frac{i\omega t}{2}}} \quad (1.13)$$